

Solutions to Theory 3: Constrained Optimization and Dual Problems

Throughout, for symmetric matrices we use

$$\langle U, V \rangle = \text{tr}(UV).$$

Task 1

We need to solve

$$\min_{x \in \mathbb{R}^n} \{c^T x : x^T A x \leq 1\}, \quad c \neq 0, \quad A \in \mathbb{S}_{++}^n.$$

Put $y = A^{1/2}x$, so that $x = A^{-1/2}y$. Then the problem becomes

$$\min_{\|y\| \leq 1} (A^{-1/2}c)^T y.$$

By the Cauchy–Schwarz inequality,

$$(A^{-1/2}c)^T y \geq -\|A^{-1/2}c\| \|y\| \geq -\|A^{-1/2}c\|.$$

Equality is attained at

$$y_* = -\frac{A^{-1/2}c}{\|A^{-1/2}c\|}.$$

Therefore

$$x_* = A^{-1/2}y_* = -\frac{A^{-1}c}{\sqrt{c^T A^{-1}c}},$$

and the optimal value is

$$p_* = -\sqrt{c^T A^{-1}c}.$$

Since $A^{-1/2}c \neq 0$, the equality case in Cauchy–Schwarz is unique; hence the minimizer is unique.

Task 2

We solve

$$\min_{X \in \mathbb{S}_{++}^n} \{\text{tr}(X^{-1}) : \langle A, X \rangle \leq b\}, \quad A \in \mathbb{S}_{++}^n, \quad b > 0.$$

The constraint is active at the optimum: if $\langle A, X \rangle < b$, then tX is still feasible for some $t > 1$, while

$$\text{tr}((tX)^{-1}) = t^{-1} \text{tr}(X^{-1}) < \text{tr}(X^{-1}).$$

The Lagrangian is

$$L(X, \lambda) = \text{tr}(X^{-1}) + \lambda(\text{tr}(AX) - b), \quad \lambda \geq 0.$$

For a symmetric perturbation H ,

$$d \text{tr}(X^{-1})[H] = -\text{tr}(X^{-1}HX^{-1}) = -\text{tr}(X^{-2}H),$$

so stationarity gives

$$-X^{-2} + \lambda A = 0, \quad \text{that is,} \quad X^{-2} = \lambda A.$$

Since X^{-1} is positive definite, the positive definite square root is unique, and hence

$$X^{-1} = \sqrt{\lambda} A^{1/2}, \quad X = \lambda^{-1/2} A^{-1/2}.$$

Using the active constraint,

$$b = \text{tr}(AX) = \lambda^{-1/2} \text{tr}(A^{1/2}),$$

therefore

$$\lambda^{1/2} = \frac{\text{tr}(A^{1/2})}{b}.$$

Thus the unique solution is

$$X_* = \frac{b}{\text{tr}(A^{1/2})} A^{-1/2},$$

and the optimal value is

$$p_* = \frac{(\text{tr}(A^{1/2}))^2}{b}.$$

Task 3

Consider

$$\max_{X \in \mathbb{S}_{++}^n} \{\det X : \|Xe_i\| \leq 1, i = 1, \dots, n\}.$$

Let $\lambda_1, \dots, \lambda_n > 0$ be the eigenvalues of X . Since $X = X^T$,

$$\sum_{i=1}^n \|Xe_i\|^2 = \sum_{i=1}^n e_i^T X^2 e_i = \text{tr}(X^2) = \sum_{i=1}^n \lambda_i^2.$$

The constraints imply $\text{tr}(X^2) \leq n$. Hence, by the arithmetic-geometric mean inequality,

$$\det X = \prod_{i=1}^n \lambda_i = \left(\prod_{i=1}^n \lambda_i^2 \right)^{1/2} \leq \left(\frac{1}{n} \sum_{i=1}^n \lambda_i^2 \right)^{n/2} \leq 1.$$

The matrix I_n is feasible and satisfies $\det I_n = 1$, so it is optimal.

If equality holds, then equality must hold in AM-GM and in $\text{tr}(X^2) \leq n$. Therefore

$$\lambda_1^2 = \dots = \lambda_n^2 = 1.$$

Since X is positive definite, $\lambda_i = 1$ for all i , and hence $X = I_n$. Thus the solution is unique and equals

$$X_* = I_n.$$

Now let $X \in \mathbb{S}_{++}^n$ be arbitrary and set

$$d_i = \|Xe_i\| > 0, \quad D = \text{diag}(d_1, \dots, d_n), \quad B = XD^{-1}.$$

Then $\|Be_i\| = 1$ for every i . Let

$$H = (B^T B)^{1/2} \in \mathbb{S}_{++}^n.$$

For every i ,

$$\|He_i\|^2 = e_i^T H^2 e_i = e_i^T B^T B e_i = \|Be_i\|^2 = 1.$$

Thus H is feasible in the maximization problem above, so $\det H \leq 1$. Since

$$\det H = \sqrt{\det(B^T B)} = |\det B|,$$

we obtain

$$|\det B| \leq 1.$$

Because X is positive definite, $\det X > 0$, and therefore

$$\det X = \det(BD) = \det B \prod_{i=1}^n d_i \leq \prod_{i=1}^n \|Xe_i\|.$$

This gives the required Hadamard inequality in the stated case:

$$\boxed{\det X \leq \|Xe_1\| \cdots \|Xe_n\|}.$$

Task 4

We solve

$$\min_{X \in \mathbb{S}_{++}^n} \{\langle C^{-1}, X \rangle - \log \det X : a^T X a \leq 1\}, \quad C \in \mathbb{S}_{++}^n, \quad a \neq 0.$$

The objective is strictly convex on $\mathbb{S}_{++}^n$, because $-\log \det X$ is strictly convex and $\langle C^{-1}, X \rangle$ is linear. The constraint is affine in X , and Slater's condition holds, for example for sufficiently small $X = \varepsilon I_n$. Hence the KKT conditions are necessary and sufficient, and strict convexity gives uniqueness.

The Lagrangian is

$$L(X, \lambda) = \langle C^{-1}, X \rangle - \log \det X + \lambda(a^T X a - 1), \quad \lambda \geq 0.$$

Stationarity gives

$$C^{-1} - X^{-1} + \lambda a a^T = 0,$$

so

$$X = (C^{-1} + \lambda a a^T)^{-1}.$$

Let

$$\gamma = a^T C a > 0.$$

By the Woodbury identity,

$$(C^{-1} + \lambda a a^T)^{-1} = C - \frac{\lambda}{1 + \lambda \gamma} C a a^T C.$$

Therefore

$$a^T X a = \gamma - \frac{\lambda \gamma^2}{1 + \lambda \gamma} = \frac{\gamma}{1 + \lambda \gamma}.$$

Complementary slackness gives

$$\lambda \left(\frac{\gamma}{1 + \lambda \gamma} - 1 \right) = 0.$$

If $\gamma \leq 1$, then $\lambda_* = 0$ is feasible and satisfies the KKT conditions. Hence

$$X_* = C.$$

If $\gamma > 1$, then the constraint is active, and

$$\frac{\gamma}{1 + \lambda \gamma} = 1.$$

Thus

$$\lambda_* = \frac{\gamma - 1}{\gamma}.$$

Substituting this value into the Woodbury formula gives

$$X_* = C - \frac{\gamma - 1}{\gamma^2} C a a^T C.$$

Combining the two cases, the solution can be written without using C^{-1} as

$$\boxed{X_* = C - \frac{(\gamma - 1)_+}{\gamma^2} C a a^T C, \quad \gamma = a^T C a,}$$

where $(t)_+ = \max\{t, 0\}$. This formula depends only on C , a , and scalar products involving them.

Task 5

The primal problem is

$$\min_{x \in \mathbb{R}^n, s \in \mathbb{R}^m} \left\{ \frac{1}{2} \|s - b\|^2 + \frac{\rho}{2} \|x\|^2 : s = Ax \right\}, \quad \rho > 0.$$

Use the equality constraint in the form $s - Ax = 0$ and introduce a multiplier $\nu \in \mathbb{R}^m$:

$$L(x, s, \nu) = \frac{1}{2} \|s - b\|^2 + \frac{\rho}{2} \|x\|^2 + \nu^T (s - Ax).$$

Minimization over s gives

$$s - b + \nu = 0, \quad s = b - \nu.$$

Minimization over x gives

$$\rho x - A^T \nu = 0, \quad x = \frac{1}{\rho} A^T \nu.$$

Substituting these stationary values into the Lagrangian yields the dual function

$$g(\nu) = b^T \nu - \frac{1}{2} \|\nu\|^2 - \frac{1}{2\rho} \|A^T \nu\|^2.$$

Therefore the dual problem is the unconstrained concave quadratic maximization problem

$$\boxed{\max_{\nu \in \mathbb{R}^m} \left\{ b^T \nu - \frac{1}{2} \|\nu\|^2 - \frac{1}{2\rho} \|A^T \nu\|^2 \right\} .}$$

Equivalently, it can be written as

$$\boxed{\min_{\nu \in \mathbb{R}^m} \left\{ \frac{1}{2} \nu^T \left(I_m + \frac{1}{\rho} A A^T \right) \nu - b^T \nu \right\} .}$$

The dual maximizer is

$$\nu_* = \left(I_m + \frac{1}{\rho} A A^T \right)^{-1} b.$$

Task 6

Let

$$\Delta_{ik} = \mathbf{1}_{\{k \neq y_i\}}.$$

The combined constraints can be written as

$$w_k^T x_i + b_k - w_{y_i}^T x_i - b_{y_i} + \Delta_{ik} - \xi_i \leq 0, \quad i = 1, \dots, N, \quad k = 1, \dots, K.$$

For $k = y_i$, this constraint is exactly $-\xi_i \leq 0$, i.e. $\xi_i \geq 0$.

Introduce Lagrange multipliers $\beta_{ik} \geq 0$. The Lagrangian is

$$\begin{aligned} L(W, b, \xi, \beta) = & \frac{1}{2} \sum_{k=1}^K \|w_k\|^2 + C \sum_{i=1}^N \xi_i \\ & + \sum_{i=1}^N \sum_{k=1}^K \beta_{ik} (w_k^T x_i + b_k - w_{y_i}^T x_i - b_{y_i} + \Delta_{ik} - \xi_i). \end{aligned}$$

Stationarity with respect to ξ_i gives

$$C - \sum_{k=1}^K \beta_{ik} = 0, \quad i = 1, \dots, N.$$

Define new variables α_{ik} by

$$\alpha_{ik} = \begin{cases} -\beta_{ik}, & k \neq y_i, \\ C - \beta_{iy_i}, & k = y_i. \end{cases}$$

Since $\sum_k \beta_{ik} = C$, this is equivalent to

$$\alpha_{iy_i} = \sum_{k \neq y_i} \beta_{ik}, \quad \alpha_{ik} = -\beta_{ik} \quad (k \neq y_i).$$

Hence

$$\sum_{k=1}^K \alpha_{ik} = 0, \quad \alpha_{ik} \leq 0 \quad (k \neq y_i), \quad \alpha_{iy_i} \leq C.$$

Moreover, $\alpha_{iy_i} \geq 0$ follows automatically from the first two conditions.

For each class ℓ , stationarity with respect to w_ℓ gives

$$w_\ell + \sum_{i=1}^N \left(\beta_{i\ell} - \mathbf{1}_{\{y_i=\ell\}} \sum_{k=1}^K \beta_{ik} \right) x_i = 0.$$

Using the definition of α , this becomes

$$w_\ell = \sum_{i=1}^N \alpha_{i\ell} x_i.$$

If the bias variables $b_1, \dots, b_K$ are free, stationarity with respect to b_ℓ also gives

$$\sum_{i=1}^N \alpha_{i\ell} = 0, \quad \ell = 1, \dots, K.$$

Thus the exact dual for the stated primal with free biases contains these additional class-balance constraints.

The linear part of the dual objective is

$$\sum_{i=1}^N \sum_{k=1}^K \beta_{ik} \Delta_{ik} = \sum_{i=1}^N \sum_{k \neq y_i} \beta_{ik} = \sum_{i=1}^N \alpha_{iy_i}.$$

The quadratic part is obtained by minimizing over the w_ℓ :

$$-\frac{1}{2} \sum_{\ell=1}^K \left\| \sum_{i=1}^N \alpha_{i\ell} x_i \right\|^2 = -\frac{1}{2} \sum_{i,j=1}^N x_i^T x_j \sum_{\ell=1}^K \alpha_{i\ell} \alpha_{j\ell}.$$

Therefore, with free biases, the dual is

$$\begin{array}{ll} \max_{\alpha} & \sum_{i=1}^N \alpha_{iy_i} - \frac{1}{2} \sum_{i,j=1}^N x_i^T x_j \sum_{k=1}^K \alpha_{ik} \alpha_{jk} \\ \text{s.t.} & \sum_{k=1}^K \alpha_{ik} = 0, & i = 1, \dots, N, \\ & \alpha_{ik} \leq 0, & i = 1, \dots, N, \ k \neq y_i, \\ & \alpha_{iy_i} \leq C, & i = 1, \dots, N, \\ & \sum_{i=1}^N \alpha_{ik} = 0, & k = 1, \dots, K. \end{array}$$

If the bias terms are fixed to zero, or if the convention is to absorb them into regularized features, the last line is absent. Then we obtain exactly the commonly printed Crammer–Singer dual

$$\begin{array}{ll} \max_{\alpha} & \sum_{i=1}^N \alpha_{iy_i} - \frac{1}{2} \sum_{i,j=1}^N x_i^T x_j \sum_{k=1}^K \alpha_{ik} \alpha_{jk} \\ \text{s.t.} & \sum_{k=1}^K \alpha_{ik} = 0, & i = 1, \dots, N, \\ & \alpha_{ik} \leq 0, & i = 1, \dots, N, \ k \neq y_i, \\ & \alpha_{iy_i} \leq C, & i = 1, \dots, N. \end{array}$$

Here the linear term is equivalently

$$\sum_{i=1}^N \alpha_{iy_i} = - \sum_{i=1}^N \sum_{k \neq y_i} \alpha_{ik},$$

because $\sum_k \alpha_{ik} = 0$.